

CoVaR[†]

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We propose a measure of systemic risk, ΔCoVaR , defined as the change in the value at risk of the financial system conditional on an institution being under distress relative to its median state. Our estimates show that characteristics such as leverage, size, maturity mismatch, and asset price booms significantly predict ΔCoVaR . We also provide out-of-sample forecasts of a countercyclical, forward-looking measure of systemic risk, and show that the 2006:IV value of this measure would have predicted more than one-third of realized ΔCoVaR during the 2007–2009 financial crisis. (JEL C58, E32, G01, G12, G17, G20, G32)

In times of financial crisis, losses spread across financial institutions, threatening the financial system as a whole.¹ The spreading of distress gives rise to systemic risk: the risk that the capacity of the entire financial system is impaired, with potentially adverse consequences for the real economy. Spillovers across institutions can occur directly due to direct contractual links and heightened counterparty credit risk or indirectly through price effects and liquidity spirals. As a result of these spillovers, the measured comovement of institutions' assets and liabilities tends to rise above and beyond levels purely justified by fundamentals. Systemic risk measures gauge the increase in tail comovement that can arise due to the spreading of financial distress across institutions.

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¹Examples include the 1987 equity market crash, which was started by portfolio hedging of pension funds and led to substantial losses of investment banks; the 1998 crisis, which was started by losses of hedge funds and spilled over to the trading floors of commercial and investment banks; and the 2007–2009 crisis, which spread from structured investment vehicles to commercial banks and on to investment banks and hedge funds. See, e.g., Brady (1988); Rubin et al. (1999); Brunnermeier (2009); and Adrian and Shin (2010b).